

Assignment 1. Creating a Business Case and Collecting Initial Data

Due Date - November 26, 2024

Evaluation - 10% of the total grade

Instructions

A major component of DIT54102 is a set of assignments where you pick and investigate applied data science problem and gain experience with the data science pipeline. This is an open-ended project, where you find a problem, collect data for analysis, understand and prepare data to the next steps of the data science pipeline. CRISP-DM should guide your data mining effort. First, research what kind of data is freely available online that may be useful. Make sure you have access to the data you want to use in the quantity and quality you need. There are thousands of publicly available datasets on diverse topics ranging from business to particle physics. Here we've enlisted some of the best sources to find publicly available datasets for your assignments project:

- **Kaggle (<https://www.kaggle.com/datasets>)** — The word “data scientist” and “Kaggle” are inextricably interlinked and everyone in the data science community is familiar with it. Kaggle is a fantastic resource for data scientists and machine learning engineers looking for datasets to work on with some pre-processing already done. It is a great place to find datasets on everything under the sun as the platform is popular for hosting multiple data science and machine learning challenges for a real problem that various organizations are trying to solve. You will find datasets of all sizes up to as large as 2TB having more than 50 million records. Users can choose from over 18,000 datasets from the Kaggle Dataset repository. One can easily find a required dataset using the search box with multiple filters such as the size of the dataset, filetype, tags, etc. One can also preview and bookmark the datasets they like.
- **Google Custom Dataset Search (<https://datasetsearch.research.google.com/>)** — The custom Google Dataset search engine was launched in September 2018. You can specify whether you are searching for data on plants, animals, diseases, UFO sighting, movies, calamities, and more. Ideally, within seconds, you should be able to find the desired dataset. It is easier to harvest datasets from the custom Google Dataset search engine using keywords, name of the dataset, creator-info, format (JSON, CSV, etc.), and description. You can also search for datasets in mark-up languages and find datasets wherever they are hosted -an author's personal page, publisher's website, or any digital library. The objective of developing this engine is to unify thousands of various dataset repositories and make that data easily available.
- **UCI Machine Learning Repository (<https://archive.ics.uci.edu/ml/datasets.php>)** — If you are looking for a dataset repository that can help you find the dataset by the type of machine learning problem, then UCI Machine learning repository is the go-to place. It has datasets classified based on characteristics such as Univariate, Time Series, Multivariate, sequential, and also based on the associated tasks such as classification, regression, clustering, etc. Most of the datasets on UCI are cleaned but they have varying levels of cleanliness because the researchers who have prepared these datasets have already done some kind of pre-processing on them like — the selection of instances and attributes. You can browse the 475 datasets based on various filters such as the number of attributes, number of instances, data type, associated tasks, attribute

type, and the subject matter. These filters can be of great help in finding the right dataset. However, if you are interested in investigating large scale problems and techniques, then this repository might not be helpful as it houses small datasets.

- **Data.gov** — Offering more than 240k datasets, the US Government's data portal hosts all sorts of amazing datasets from climate to crime. Data.gov is an aggregator of publicly available free data from various US government agencies. Anyone can easily download the data but much of the data on this website requires research making it difficult to figure out the right version of the dataset.

For this assignment, we will go through “Business Understanding” phase. First step is to formulate a problem that you would like to solve using data. For this assignment produce an APA formatted 2-3 pages report (docx or pdf) that includes the following section:

Business/Research Objectives

Give background on the problem you plan to solve. Gather background information about the current business/research situation and document motivation and objective(s). Determine data mining success from a business/research perspective.

Data Mining Goals

Next describe what kind of problem you are trying to solve from data mining or machine learning perspective. Use CRISM-DM for guidance. List data sources that you plan to use, provide descriptions and references.

Project Plan

Describe the intended plan for achieving the data mining goals and thereby achieving the business goals. The plan should specify the steps to be performed during the rest of the project.

Scoring Rubric

Category	Criteria	Maximum (Points)
Business Objectives	Background, current situation, and motivation explained.	2
Data Mining Goals	Business/research questions stated clearly.	
Project Plan	Problem type from ML perspective is determined. Potential data resources are listed, researched, and clearly explained.	2
Writing and Readability	Complete.	1
Originality	Report is exceptionally well organized and well written, with charts and tables embedded in report.	3
	An original research topic was selected for this project.	2
Total		10